

ASP.NET Lab Programs

Program 1: Exposure of Web Applications and Tools

Aim: To demonstrate ASP.NET project structure and tools.

Program1.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="Program1.aspx.cs"
Inherits="Program1" %>

<!DOCTYPE html>

<html>

<head>

    <title>Program 1 - Exposure of Web Applications</title>

</head>

<body>

    <form id="form1" runat="server">

        <h2>Welcome to ASP.NET Web Application</h2>

        <asp:Label ID="lblInfo" runat="server" Font-Size="Large"></asp:Label>

    </form>

</body>

</html>
```

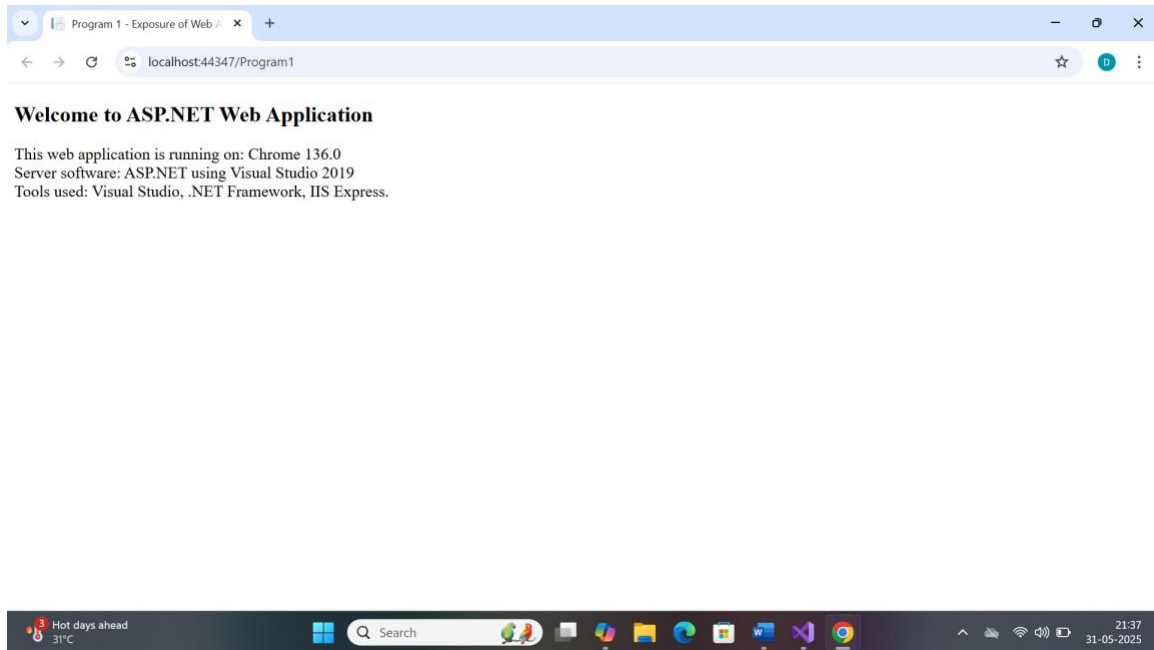
Program1.aspx.cs:

```
using System;

public partial class Program1 : System.Web.UI.Page
{
    protected void Page_Load(object sender, EventArgs e)
    {
        lblInfo.Text = "This web application is running on: " +
Request.Browser.Browser + " " + Request.Browser.Version + "<br />Server
software: ASP.NET using Visual Studio 2019<br />" +
        "Tools used: Visual Studio, .NET Framework, IIS Express.";
    }
}
```

Output:

A web page displaying the heading and description of project structure and tools.



Program 2: Implement HTML Controls

Use HTML controls and access their values in code-behind.

Program2.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="Program2.aspx.cs"
Inherits="Program2" %>

<!DOCTYPE html>

<html>

<head><title>Program 2 - HTML Controls</title></head>

<body>

    <form id="form1" runat="server">

        <label for="name">Enter your name:</label>

        <input type="text" name="name" />

        <br /><br />

        <input type="submit" value="Submit" />

    </form>

</body>

</html>
```

Program2.aspx.cs:

```
using System;

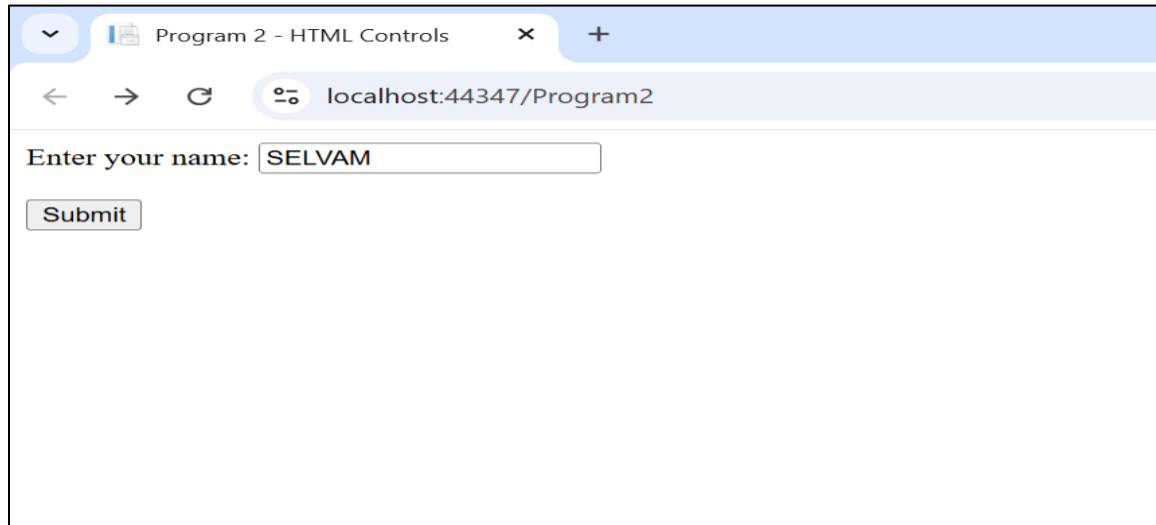
public partial class Program2 : System.Web.UI.Page
{
    protected void Page_Load(object sender, EventArgs e)
    {
        if (IsPostBack)
        {
            string name = Request.Form["name"];

            Response.Write("Hello, " + name);
        }
    }
}
```

```
}  
}
```

Output:

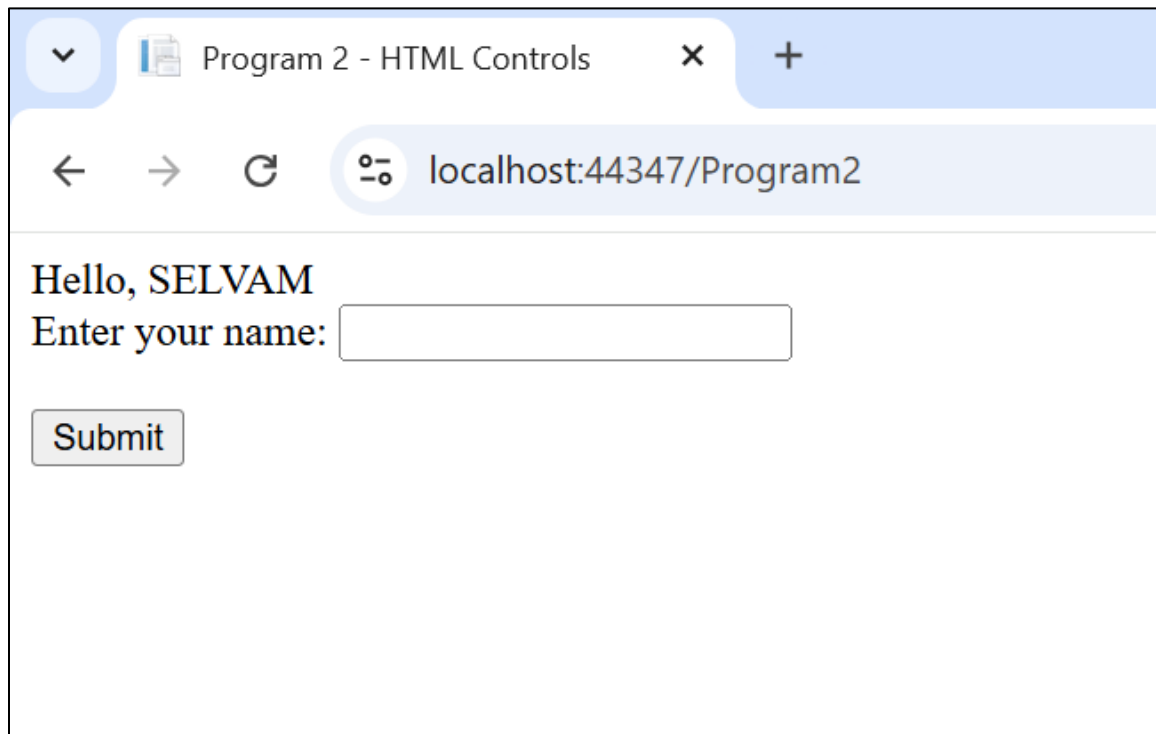
When user enters a name and clicks Show Name, the page displays "Hello, [Name]!".



Program 2 - HTML Controls

localhost:44347/Program2

Enter your name:



Program 2 - HTML Controls

localhost:44347/Program2

Hello, SELVAM

Enter your name:

Program 3: Implement Server Controls

Use ASP.NET server controls (TextBox, Button, Label) to get and display input.

Program3.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="Program3.aspx.cs"
Inherits="Program3" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title>Program 3</title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <asp:TextBox ID="txtInput" runat="server" Placeholder="Enter
Text"></asp:TextBox><br /><br />
            <asp:Button ID="btnDisplay" runat="server" Text="Display"
OnClick="btnDisplay_Click" /><br /><br />
            <asp:Label ID="lblOutput" runat="server"></asp:Label>
        </div>
    </form>
</body>
</html>
```

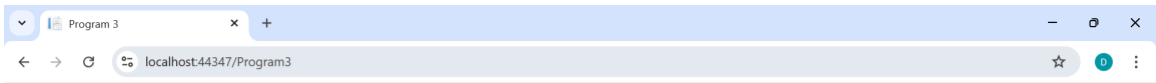
Program3.aspx.cs:

```
using System;

public partial class Program3 : System.Web.UI.Page
{
    protected void btnDisplay_Click(object sender, EventArgs e)
    {
        lblOutput.Text = "You entered: " + txtInput.Text;
    }
}
```

Output:

On clicking Display, label shows "You entered: [InputText]".



WELCOME TO ASP LAB

Display

You entered: WELCOME TO ASP LAB



Program 4: Web Application Using Web Controls

Demonstrate various web controls like DropDownList, CheckBox, RadioButton, and Buttons.

Program4.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="Program4.aspx.cs"
Inherits="Program4" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title>Program 4</title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <asp:DropDownList ID="ddlColors" runat="server">
                <asp:ListItem Text="Red" Value="Red"></asp:ListItem>
                <asp:ListItem Text="Green" Value="Green"></asp:ListItem>
                <asp:ListItem Text="Blue" Value="Blue"></asp:ListItem>
            </asp:DropDownList><br /><br />
            <asp:CheckBox ID="chkBold" runat="server" Text="Bold" /><br />
            <asp:RadioButtonList ID="rblSize" runat="server">
                <asp:ListItem Text="10" Value="10"></asp:ListItem>
                <asp:ListItem Text="14" Value="14"></asp:ListItem>
                <asp:ListItem Text="18" Value="18"></asp:ListItem>
            </asp:RadioButtonList><br /><br />
            <asp:Button ID="btnStyle" runat="server" Text="Apply Style"
OnClick="btnStyle_Click" /><br /><br />
            <asp:Label ID="lblStyledText" runat="server" Text="Sample
Text"></asp:Label>
        </div>
    </form>
</body>
</html>
```

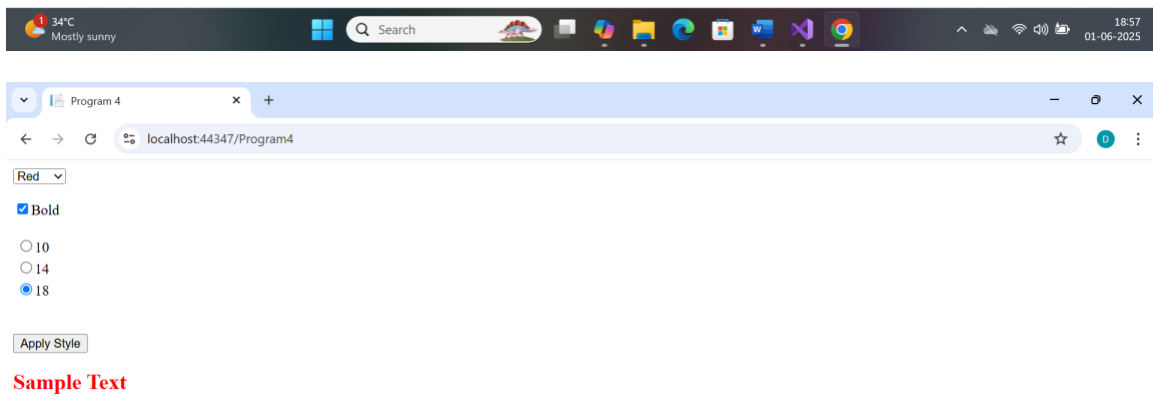
Program4.aspx.cs:

```
using System;

public partial class Program4 : System.Web.UI.Page
{
    protected void btnStyle_Click(object sender, EventArgs e)
    {
        lblStyledText.ForeColor =
System.Drawing.Color.FromName(ddlColors.SelectedValue);
        lblStyledText.Font.Bold = chkBold.Checked;
        lblStyledText.Font.Size =
FontUnit.Point(Convert.ToInt32(rblSize.SelectedValue));
    }
}
```

Output:

Changes label color, boldness, and font size based on selected controls.



Program 5: Web Application Using List Controls

Show ListBox and CheckBoxList controls and display selected items.

Program5.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="Program5.aspx.cs"
Inherits="Program5" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title>Program 5</title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <asp:ListBox ID="lstFruits" runat="server"
SelectionMode="Multiple">
                <asp:ListItem>Apple</asp:ListItem>
                <asp:ListItem>Banana</asp:ListItem>
                <asp:ListItem>Cherry</asp:ListItem>
            </asp:ListBox><br /><br />
            <asp:CheckBoxList ID="cblHobbies" runat="server">
                <asp:ListItem>Reading</asp:ListItem>
                <asp:ListItem>Traveling</asp:ListItem>
                <asp:ListItem>Swimming</asp:ListItem>
            </asp:CheckBoxList><br /><br />
            <asp:Button ID="btnShowSelections" runat="server" Text="Show
Selections" OnClick="btnShowSelections_Click" /><br /><br />
            <asp:Label ID="lblSelections" runat="server"></asp:Label>
        </div>
    </form>
</body>
</html>
```

Program5.aspx.cs:

```
using System;
using System.Text;

public partial class Program5 : System.Web.UI.Page
{
    protected void btnShowSelections_Click(object sender, EventArgs e)
    {
        StringBuilder sb = new StringBuilder();
        sb.Append("Fruits: ");
        foreach (System.Web.UI.WebControls.ListItem li in lstFruits.Items)
        {
            if (li.Selected)
                sb.Append(li.Text + ", ");
        }
        sb.Append(" | Hobbies: ");
        foreach (System.Web.UI.WebControls.ListItem li in cblHobbies.Items)
        {
            if (li.Selected)
```

```

        sb.Append(li.Text + ", ");
    }
    lblSelections.Text = sb.ToString();
}
}

```

Output:

Displays selected fruits and hobbies separated by commas.



Program 6: Web Page Design Using Rich Controls and Validation Controls; File Handling

Use Calendar control, FileUpload, and Validation controls; implement file upload. (Create a folder Uploads in project by right click the project click add folder give name Uploads)

Program6.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="Program6.aspx.cs"
Inherits="Program6" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title>Program 6</title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <asp:Calendar ID="Calendar1" runat="server"
OnSelectionChanged="Calendar1_SelectionChanged"></asp:Calendar><br /><br />
            Select Date: <asp:TextBox ID="txtDate" runat="server"
ReadOnly="True"></asp:TextBox><br /><br />
            <asp:RequiredFieldValidator ControlToValidate="fuFile"
ErrorMessage="Select a file"
runat="server"></asp:RequiredFieldValidator><br /><br />
            <asp:FileUpload ID="fuFile" runat="server" /><br /><br />
            <asp:Button ID="btnSubmit" runat="server" Text="Submit"
OnClick="btnSubmit_Click" /><br /><br />
            <asp:Label ID="lblMessage" runat="server"></asp:Label>
        </div>
    </form>
</body>
</html>
```

Program6.aspx.cs:

```
using System;

public partial class Program6 : System.Web.UI.Page
{
    protected void Page_Load(object sender, EventArgs e)
    {
        if (!IsPostBack)
        {
            txtDate.Text = DateTime.Now.ToShortDateString();
        }
    }

    protected void Calendar1_SelectionChanged(object sender, EventArgs e)
    {
        txtDate.Text = Calendar1.SelectedDate.ToShortDateString();
    }

    protected void btnSubmit_Click(object sender, EventArgs e)
```

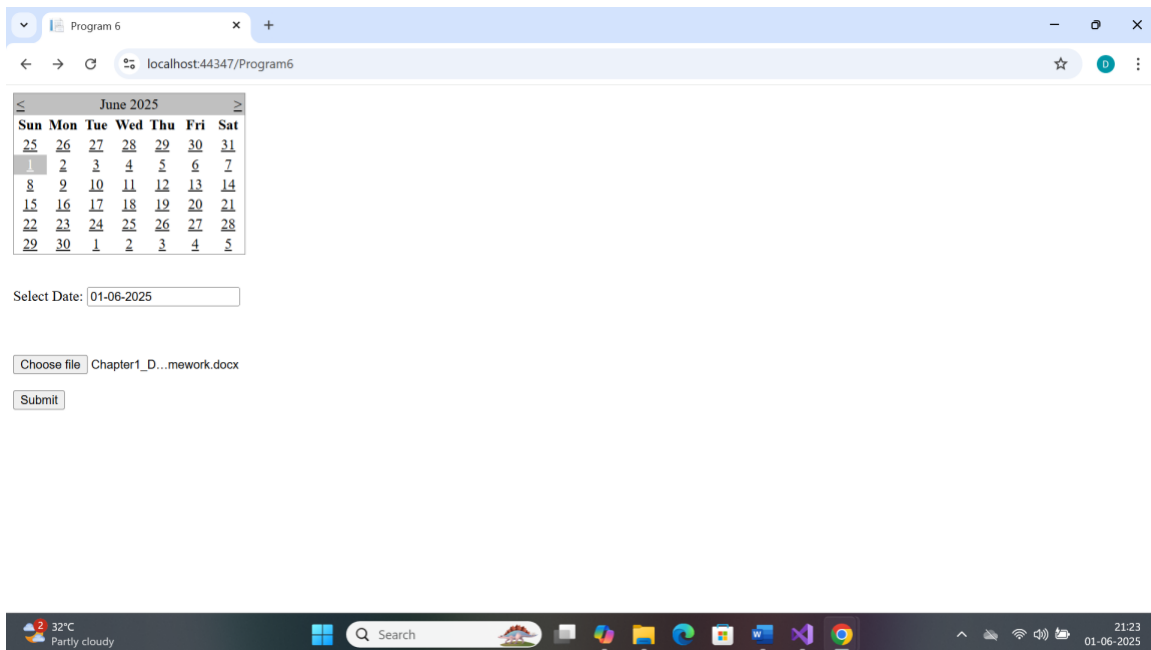
```

    {
        if (fuFile.HasFile)
        {
            string path = Server.MapPath("~/Uploads/") + fuFile.FileName;
            fuFile.SaveAs(path);
            lblMessage.Text = "Date: " + txtDate.Text + " | File saved at:
" + path;
        }
        else
        {
            lblMessage.Text = "Please select a file.";
        }
    }
}

```

Output:

Calendar displays current date; on selecting a date, txtDate updates. On selecting a file and clicking Submit, displays date and server path of saved file.



June 2025						
Sun	Mon	Tue	Wed	Thu	Fri	Sat
25	26	27	28	29	30	31
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	1	2	3	4	5

Select Date: 01-06-2025

Choose file No file chosen

Submit

Date: 01-06-2025 | File saved at: E:\asp\WebApplication1\Uploads\Chapter1_DotNet_Framework.docx

Program 7: Web Application Using Data Controls

Bind a GridView to a SqlDataSource and display data from database.

Program7.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="gridview.aspx.cs"
Inherits="GridView1.gridview" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <asp:GridView runat="server" AutoGenerateColumns="False"
DataKeyNames="StudentId" DataSourceID="SqlDataSource1">
                <Columns>
                    <asp:BoundField DataField="StudentId" HeaderText="StudentId" InsertVisible="False"
ReadOnly="True" SortExpression="StudentId" />
                    <asp:BoundField DataField="SName" HeaderText="SName" SortExpression="SName"
/>
                </Columns>
            </asp:GridView>

            <asp:SqlDataSource ID="SqlDataSource1" runat="server" ConnectionString="<%%$
ConnectionStrings:SchoolDBConnectionString %>" SelectCommand="SELECT * FROM
[Student]"></asp:SqlDataSource>
```

```

        </div>

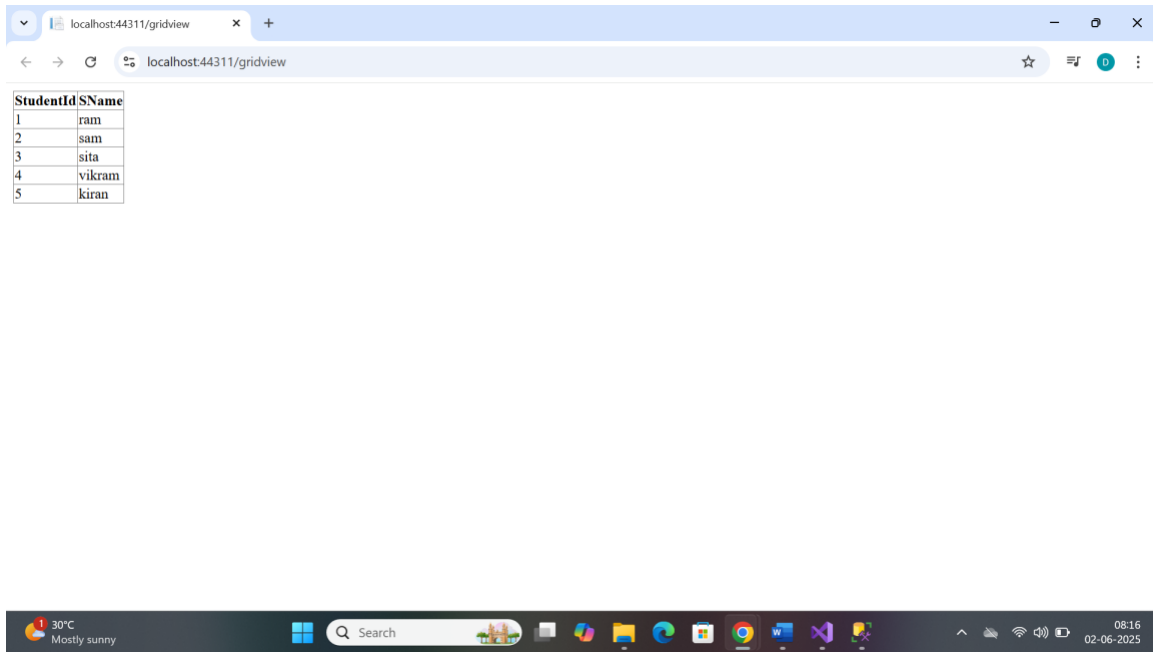
    </form>

</body>

</html>

```

Output:



Program 8: Data Binding with Web Controls

Bind DropDownList to database data from SqlDataSource.

Program8.aspx:

```

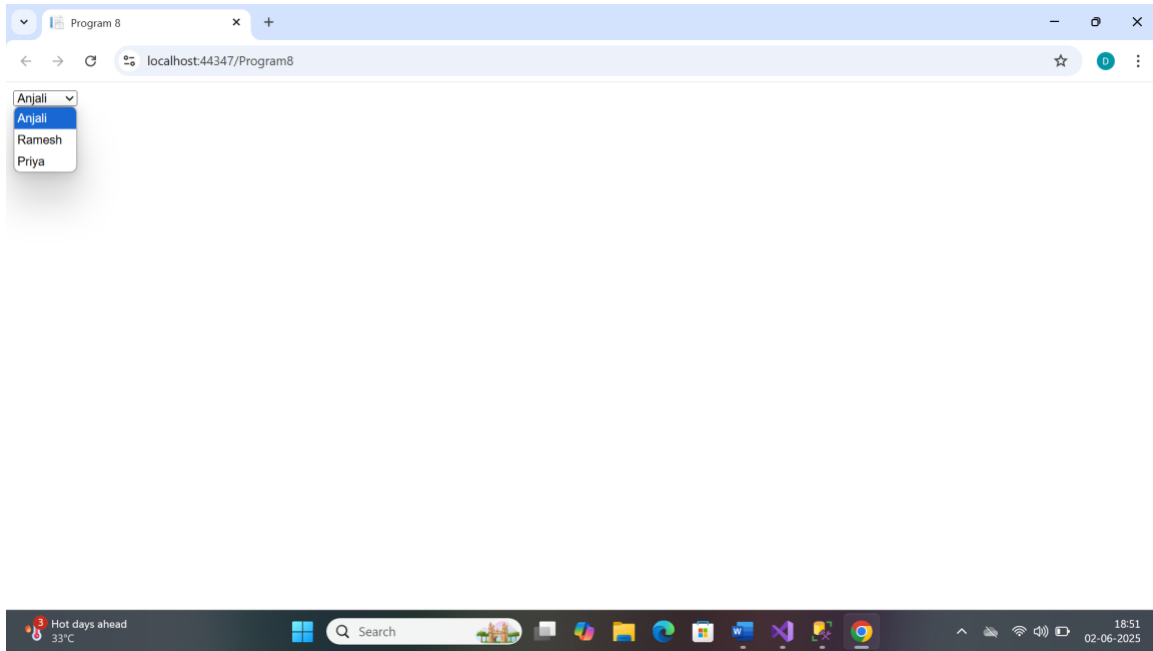
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="Program8.aspx.cs"
Inherits="Program8" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title>Program 8</title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <asp:DropDownList ID="ddlStudents" runat="server"
DataSourceID="SqlDataSource1" DataTextField="Name"
DataValueField="StudentID"></asp:DropDownList>
            <asp:SqlDataSource ID="SqlDataSource1" runat="server"
ConnectionString="<%$ ConnectionStrings:SchoolDBConnectionString %>"

```

```
SelectCommand="SELECT StudentID, Name FROM Students"></asp:SqlDataSource>
</div>
</form>
</body>
</html>
```

Output:

DropDownList populated with student names from database; selecting an item sets
SelectedValue to StudentID.



Program 9: Data Binding with Data Controls

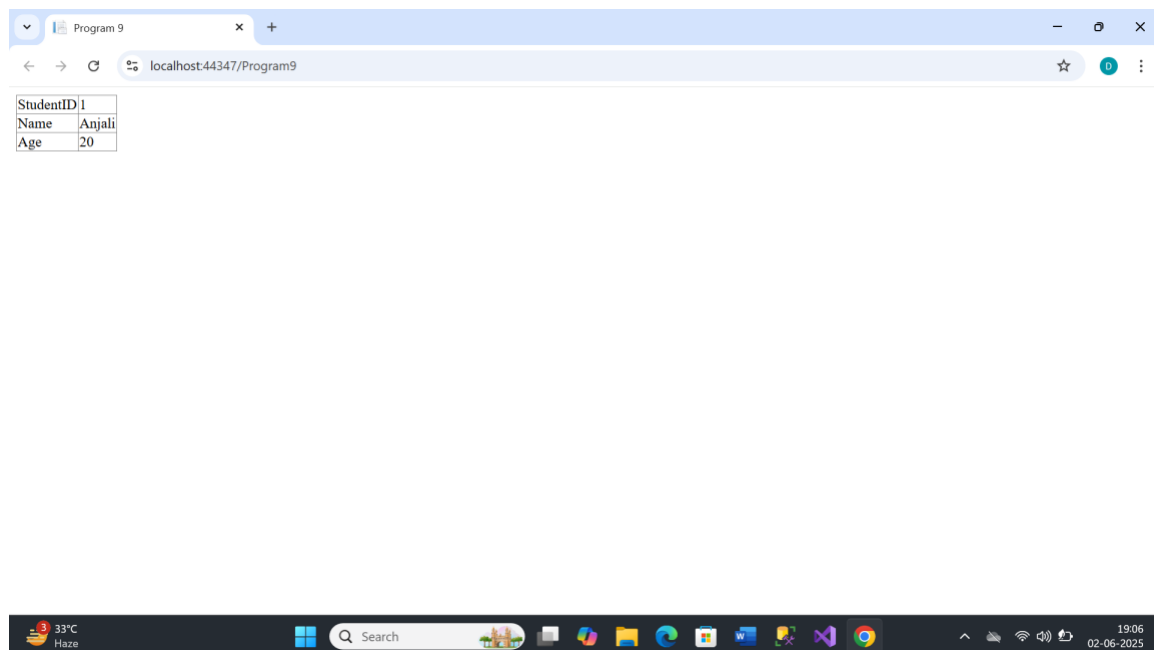
Use DetailsView to display single record and bind to SqlDataSource.

Program9.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="Program9.aspx.cs"
Inherits="Program9" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title>Program 9</title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <asp:DetailsView ID="DetailsView1" runat="server"
AutoGenerateRows="True" DataKeyNames="StudentID"
DataSourceID="SqlDataSource1"></asp:DetailsView>
            <asp:SqlDataSource ID="SqlDataSource1" runat="server"
ConnectionString="<%$ ConnectionStrings:SchoolDBConnectionString %>"
SelectCommand="SELECT StudentID, Name, Age FROM Students WHERE
StudentID=1"></asp:SqlDataSource>
        </div>
    </form>
</body>
</html>
```

Output:

DetailsView displays the record of student with StudentID = 1.



Program 10: Database Application to Perform Insert, Update, Delete Operations

Use SqlCommand in code-behind to perform CRUD operations via buttons.

Program10.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="Program10.aspx.cs"
Inherits="Program10" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">

<head runat="server">

    <title>Program 10 - Insert, Update, Delete with Auto ID</title>

</head>

<body>

    <form id="form1" runat="server">

        <div>

            Student ID:<br />

            <asp:TextBox ID="txtID" runat="server" ReadOnly="true" /><br /><br />

            Name:<br />

            <asp:TextBox ID="txtName10" runat="server" /><br /><br />

            Age:<br />

            <asp:TextBox ID="txtAge10" runat="server" /><br /><br />

            <asp:Button ID="btnInsert" runat="server" Text="Insert"
OnClick="btnInsert_Click" />

            <asp:Button ID="btnUpdate" runat="server" Text="Update"
OnClick="btnUpdate_Click" />

            <asp:Button ID="btnDelete" runat="server" Text="Delete"
OnClick="btnDelete_Click" /><br /><br />

        </div>

    </form>

</body>

</html>
```

```
        <asp:Label ID="lblResult10" runat="server" ForeColor="Green" /><br /><br />
```

```
        <asp:GridView ID="GridView1" runat="server"
AutoGenerateColumns="True"
OnSelectedIndexChanged="GridView1_SelectedIndexChanged"
AutoGenerateSelectButton="True">
```

```
        </asp:GridView>
```

```
    </div>
```

```
    </form>
```

```
</body>
```

```
</html>
```

Program10.aspx.cs:

```
using System;
```

```
using System.Data;
```

```
using System.Data.SqlClient;
```

```
using System.Web.UI.WebControls;
```

```
public partial class Program10 : System.Web.UI.Page
```

```
{
```

```
    string connString =
System.Configuration.ConfigurationManager.ConnectionStrings["SchoolDBConnectio
nString2"].ConnectionString;
```

```
    protected void Page_Load(object sender, EventArgs e)
```

```
    {
```

```
        if (!IsPostBack)
```

```
        {
```

```
            LoadGrid();
```

```

    }
}

private void LoadGrid()
{
    using (SqlConnection conn = new SqlConnection(connString))
    {
        string sql = "SELECT * FROM Students";
        SqlDataAdapter da = new SqlDataAdapter(sql, conn);
        DataTable dt = new DataTable();
        da.Fill(dt);
        GridView1.DataSource = dt;
        GridView1.DataBind();
    }
}

protected void btnInsert_Click(object sender, EventArgs e)
{
    using (SqlConnection conn = new SqlConnection(connString))
    {
        conn.Open();

        string sql = "INSERT INTO Students (Name, Age) OUTPUT  

INSERTED.StudentID VALUES (@Name, @Age)";

        using (SqlCommand cmd = new SqlCommand(sql, conn))
        {
            cmd.Parameters.AddWithValue("@Name", txtName10.Text);
            cmd.Parameters.AddWithValue("@Age", txtAge10.Text);
            int newID = (int)cmd.ExecuteScalar();

```

```

        txtID.Text = newID.ToString();

        lblResult10.Text = "Record inserted with ID: " + newID;
    }

}

LoadGrid();
}

```

```

protected void btnUpdate_Click(object sender, EventArgs e)
{
    if (string.IsNullOrEmpty(txtID.Text))
    {
        lblResult10.Text = "Select a record to update.";
        return;
    }

    using (SqlConnection conn = new SqlConnection(connString))
    {
        conn.Open();

        string sql = "UPDATE Students SET Name=@Name, Age=@Age WHERE
StudentID=@ID";

        using (SqlCommand cmd = new SqlCommand(sql, conn))
        {
            cmd.Parameters.AddWithValue("@Name", txtName10.Text);
            cmd.Parameters.AddWithValue("@Age", txtAge10.Text);
            cmd.Parameters.AddWithValue("@ID", txtID.Text);

            int rows = cmd.ExecuteNonQuery();

            lblResult10.Text = rows + " row(s) updated.";
        }
    }
}

```

```
    }  
    LoadGrid();  
}
```

```
protected void btnDelete_Click(object sender, EventArgs e)  
{
```

```
    if (string.IsNullOrEmpty(txtID.Text))  
    {  
        lblResult10.Text = "Select a record to delete.";  
        return;  
    }
```

```
    using (SqlConnection conn = new SqlConnection(connString))  
    {  
        conn.Open();  
        string sql = "DELETE FROM Students WHERE StudentID=@ID";  
        using (SqlCommand cmd = new SqlCommand(sql, conn))  
        {  
            cmd.Parameters.AddWithValue("@ID", txtID.Text);  
            int rows = cmd.ExecuteNonQuery();  
            lblResult10.Text = rows + " row(s) deleted.";  
        }  
    }
```

```
    ClearForm();  
    LoadGrid();  
}
```

```
protected void GridView1_SelectedIndexChanged(object sender, EventArgs e)
```

```

{
    GridViewRow row = GridView1.SelectedRow;

    txtID.Text = row.Cells[1].Text;

    txtName10.Text = row.Cells[2].Text;

    txtAge10.Text = row.Cells[3].Text;
}

private void ClearForm()
{
    txtID.Text = "";

    txtName10.Text = "";

    txtAge10.Text = "";
}
}

```

Output:

Insert, Update, Delete or View operations performed on Students table. lblResult10 displays number of affected rows.

Student ID:

Name:

Age:

	StudentId	Name	Age
Select	1	Anjali	20
Select	2	Ramesh	21
Select	3	Priya	19

Program 10 - Insert, Update, Delete

localhost:44347/Program10

Student ID:

4

Name:

Raghu

Age:

18

Insert

Update

Delete

Record inserted with ID: 4

	StudentId	Name	Age
Select	1	Anjali	20
Select	2	Ramesh	21
Select	3	Priya	19
Select	4	Raghu	18



Program 10 - Insert, Update, Delete

localhost:44347/Program10

Student ID:

4

Name:

Raghu Ram

Age:

22

Insert

Update

Delete

1 row(s) updated.

	StudentId	Name	Age
Select	1	Anjali	20
Select	2	Ramesh	21
Select	3	Priya	19
Select	4	Raghu Ram	22



Student ID:

Name:

Age:

1 row(s) deleted.

	StudentId	Name	Age
Select	1	Anjali	20
Select	2	Ramesh	21
Select	3	Priya	19

Program 11: Database Application Using Data Controls for CRUD with Paging and Sorting

Use GridView bound to SqlDataSource with Edit, Delete, Paging, Sorting enabled.

Program11.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="Program11.aspx.cs"
Inherits="Program11" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">

<head runat="server">

    <title>Program 11</title>

</head>

<body>

    <form id="form2" runat="server">

        <div>

            <h3>Insert New Student</h3>

            <table>

                <tr>

                    <td>Name:</td>

                    <td><asp:TextBox ID="txtName11" runat="server" /></td>

                </tr>

                <tr>

                    <td>Age:</td>

                    <td><asp:TextBox ID="txtAge11" runat="server" /></td>

                </tr>

                <tr>

                    <td colspan="2">

                        <asp:Button ID="btnInsert11" runat="server"
Text="Insert" OnClick="btnInsert11_Click" />

                    </td>

                </tr>

            </table>

        </div>

    </form>

</body>

</html>
```

```

        </tr>

</table>

<asp:Label ID="lblMessage11" runat="server" ForeColor="Green" />

<br /><br />

<asp:GridView ID="GridView2" runat="server"
AutoGenerateColumns="False"

    DataKeyNames="StudentID" DataSourceID="SqlDataSource1"

    AllowPaging="True" AllowSorting="True" PageSize="5">

    <Columns>

        <asp:BoundField DataField="StudentID" HeaderText="ID"
ReadOnly="True" SortExpression="StudentID" />

        <asp:BoundField DataField="Name" HeaderText="Name"
SortExpression="Name" />

        <asp:BoundField DataField="Age" HeaderText="Age"
SortExpression="Age" />

        <asp:CommandField ShowEditButton="True"
ShowDeleteButton="True" />

    </Columns>

</asp:GridView>

<asp:SqlDataSource ID="SqlDataSource1" runat="server"

    ConnectionString="<%=
ConnectionStrings:SchoolDBConnectionString2 %>"

    SelectCommand="SELECT StudentID, Name, Age FROM Students"

    UpdateCommand="UPDATE Students SET Name=@Name, Age=@Age WHERE
StudentID=@StudentID"

    DeleteCommand="DELETE FROM Students WHERE
StudentID=@StudentID">

    <UpdateParameters>

        <asp:Parameter Name="Name" Type="String" />

```

```

        <asp:Parameter Name="Age" Type="Int32" />
        <asp:Parameter Name="StudentID" Type="Int32" />
    </UpdateParameters>

    <DeleteParameters>

        <asp:Parameter Name="StudentID" Type="Int32" />

    </DeleteParameters>

</asp:SqlDataSource>

</div>

</form>

</body>

</html>

```

Program11.aspx.cs:

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
using System.Data.SqlClient;

public partial class Program11 : System.Web.UI.Page
{
    string connString =
System.Configuration.ConfigurationManager.ConnectionStrings["SchoolDBConnectio
nString2"].ConnectionString;

    protected void Page_Load(object sender, EventArgs e)
    {

```

```
}
```

```
protected void btnInsert11_Click(object sender, EventArgs e)
```

```
{
```

```
    if (txtName11.Text.Trim() == "" || txtAge11.Text.Trim() == "")
```

```
    {
```

```
        lblMessage11.Text = "Please enter Name and Age.";
```

```
        lblMessage11.ForeColor = System.Drawing.Color.Red;
```

```
        return;
```

```
    }
```

```
    using (SqlConnection conn = new SqlConnection(connString))
```

```
    {
```

```
        conn.Open();
```

```
        string sql = "INSERT INTO Students (Name, Age) VALUES (@Name,  
@Age)";
```

```
        using (SqlCommand cmd = new SqlCommand(sql, conn))
```

```
        {
```

```
            cmd.Parameters.AddWithValue("@Name", txtName11.Text.Trim());
```

```
            cmd.Parameters.AddWithValue("@Age", txtAge11.Text.Trim());
```

```
            int rows = cmd.ExecuteNonQuery();
```

```
            lblMessage11.Text = $"{rows} row(s) inserted.";
```

```
            lblMessage11.ForeColor = System.Drawing.Color.Green;
```

```
        }
```

```
    }
```

```
    txtName11.Text = "";
```

```
    txtAge11.Text = "";
```

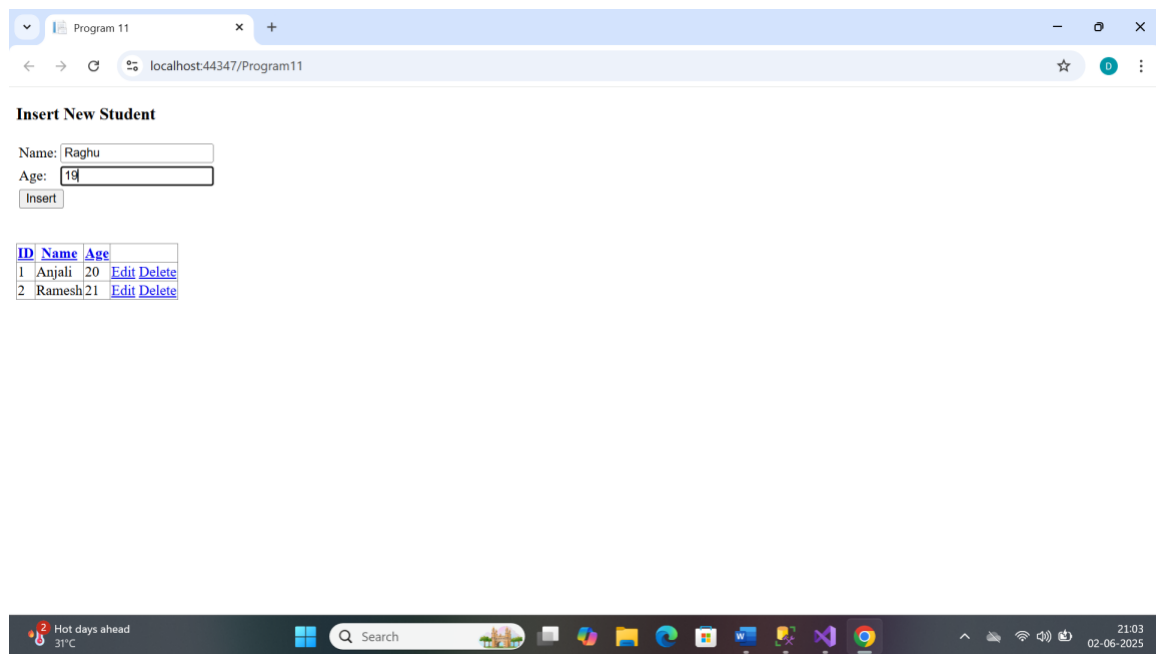
```

        GridView2.DataBind(); // refresh grid
    }
}

```

Output:

GridView displays Students data with Edit, Delete buttons, and supports paging(more than 5 records display in next page) and sorting (ID, Name, Age).



Program 11

localhost:44347/Program11

Insert New Student

Name:

Age:

Insert

1 row(s) inserted.

ID	Name	Age	
1	Anjali	20	Edit Delete
2	Ramesh	21	Edit Delete
5	Raghu	19	Edit Delete

Hot days ahead
31°C

Search

21:04
02-06-2025

Program 11

localhost:44347/Program11

Insert New Student

Name:

Age:

Insert

1 row(s) inserted.

ID	Name	Age	
1	Anjali	20	Edit Delete
2	Ramesh	21	Edit Delete
5	Raghu	19	Edit Delete
6	Priya	21	Edit Delete
7	Ravi	18	Edit Delete

1

[2](#)



Insert New Student

Name:

Age:

1 row(s) inserted.

ID	Name	Age	
8	Balu	22	Edit Delete
9	Helen	21	Edit Delete
1	2		



Insert New Student

Name:

Age:

1 row(s) inserted.

ID	Name	Age	
7	Ravi	18	Edit Delete
5	Raghu	19	Edit Delete
1	Anjali	20	Edit Delete
2	Ramesh	21	Edit Delete
6	Priya	21	Edit Delete
1	2		



Program 11

localhost:44347/Program11

Insert New Student

Name:

Age:

Insert

1 row(s) inserted.

ID	Name	Age	
7	Ravi	18	Edit Delete
5	Raghu	19	Edit Delete
1	Anjali	20	Edit Delete
2	Ramesh	21	Edit Delete
6	Priya Ravi	21	Update Cancel

1 [2](#)

31°C
Partly cloudy

Search

21:08

02-06-2025

Program 11

localhost:44347/Program11

Insert New Student

Name:

Age:

Insert

1 row(s) inserted.

ID	Name	Age	
7	Ravi	18	Edit Delete
5	Raghu	19	Edit Delete
1	Anjali	20	Edit Delete
2	Ramesh	21	Edit Delete
6	Priya Ravi	21	Edit Delete

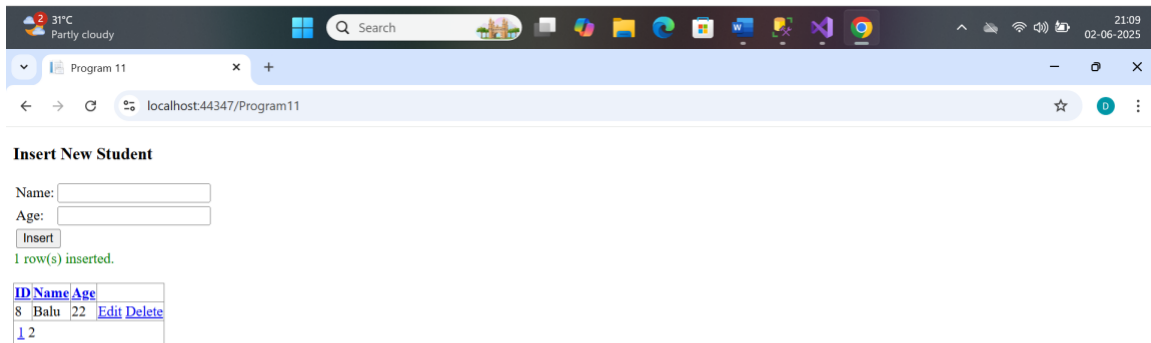
1 [2](#)

31°C
Partly cloudy

Search

21:08

02-06-2025



Program 12: Implement XML Classes

Use XmlDocument to read and XmlTextWriter to write XML.

Program12.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="Program12.aspx.cs"
Inherits="Program12" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
```

```

        <title>Program 12</title>
    </head>
    <body>
        <form id="form1" runat="server">
            <div>
                <asp:Button ID="btnRead" runat="server" Text="Read XML"
OnClick="btnRead_Click" /><br /><br />
                <asp:GridView ID="GridView1" runat="server"
AutoGenerateColumns="True"></asp:GridView><br /><br />
                ID: <asp:TextBox ID="txtID12" runat="server"></asp:TextBox><br
/><br />
                Name: <asp:TextBox ID="txtName12"
runat="server"></asp:TextBox><br /><br />
                Age: <asp:TextBox ID="txtAge12"
runat="server"></asp:TextBox><br /><br />
                <asp:Button ID="btnWrite" runat="server" Text="Write XML"
OnClick="btnWrite_Click" /><br /><br />
                <asp:Label ID="lblStatus12" runat="server"></asp:Label>
            </div>
        </form>
    </body>
</html>

```

Program12.aspx.cs:

```
using System;
```

```
using System.Data;
```

```
using System.Xml;
```

```
public partial class Program12 : System.Web.UI.Page
```

```
{
```

```
    protected void btnRead_Click(object sender, EventArgs e)
```

```
    {
```

```
        XmlDocument xmlDoc = new XmlDocument();
```

```
        xmlDoc.Load(Server.MapPath("~/App_Data/NewStudents.xml"));
```

```
        XmlNodeList nodes = xmlDoc.SelectNodes("/Students/Student");
```

```
        DataTable dt = new DataTable();
```

```
        dt.Columns.Add("StudentID");
```

```
        dt.Columns.Add("Name");
```

```

dt.Columns.Add("Age");

foreach (XmlNode node in nodes)
{
    DataRow dr = dt.NewRow();
    dr["StudentID"] = node["StudentID"].InnerText;
    dr["Name"] = node["Name"].InnerText;
    dr["Age"] = node["Age"].InnerText;
    dt.Rows.Add(dr);
}
GridView1.DataSource = dt;
GridView1.DataBind();
}

protected void btnWrite_Click(object sender, EventArgs e)
{
    string filePath = Server.MapPath("~/App_Data/NewStudents.xml");

    XmlDocument xmlDoc = new XmlDocument();

    // If file exists, load and append
    if (System.IO.File.Exists(filePath))
    {
        xmlDoc.Load(filePath);

        XmlNode root = xmlDoc.DocumentElement;

        XmlElement student = xmlDoc.CreateElement("Student");

```

```

XmlElement id = xmlDoc.CreateElement("StudentID");

id.InnerText = txtID12.Text;

student.AppendChild(id);


XmlElement name = xmlDoc.CreateElement("Name");

name.InnerText = txtName12.Text;

student.AppendChild(name);


XmlElement age = xmlDoc.CreateElement("Age");

age.InnerText = txtAge12.Text;

student.AppendChild(age);


root.AppendChild(student);
}
else
{
    // If file doesn't exist, create new structure

    XmlDeclaration xmlDecl = xmlDoc.CreateXmlDeclaration("1.0",
"UTF-8", null);

    xmlDoc.AppendChild(xmlDecl);


    XmlElement root = xmlDoc.CreateElement("Students");

    xmlDoc.AppendChild(root);


    XmlElement student = xmlDoc.CreateElement("Student");


    XmlElement id = xmlDoc.CreateElement("StudentID");

    id.InnerText = txtID12.Text;

```

```

        student.AppendChild(id);

        XmlElement name = xmlDoc.CreateElement("Name");
        name.InnerText = txtName12.Text;
        student.AppendChild(name);

        XmlElement age = xmlDoc.CreateElement("Age");
        age.InnerText = txtAge12.Text;
        student.AppendChild(age);

        root.AppendChild(student);
    }

    xmlDoc.Save(filePath);

    lblStatus12.Text = "Student added to XML file.";
}
}

```

Output:

Enter ID, Name, Age and click Write XML to create NewStudents.xml and display success message. Click Read XML to populate GridView from Students.xml.

Program 12

localhost:44347/Program12

Read XML

ID:

Name:

Age:

Write XML



Program 12

localhost:44347/Program12

Read XML

ID:

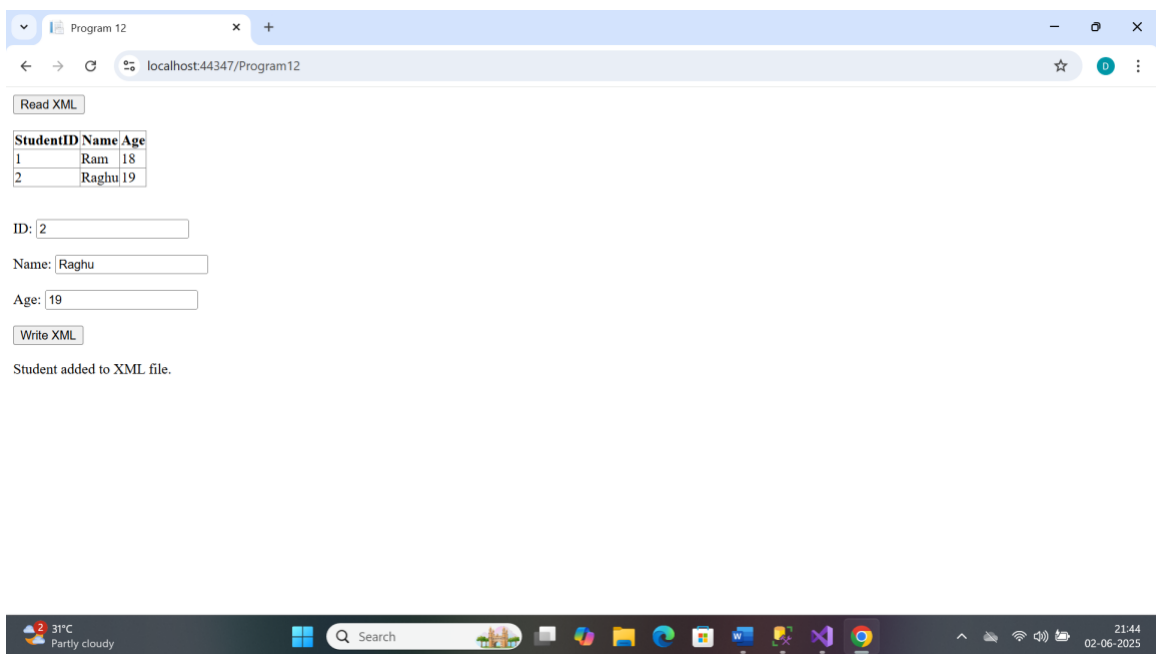
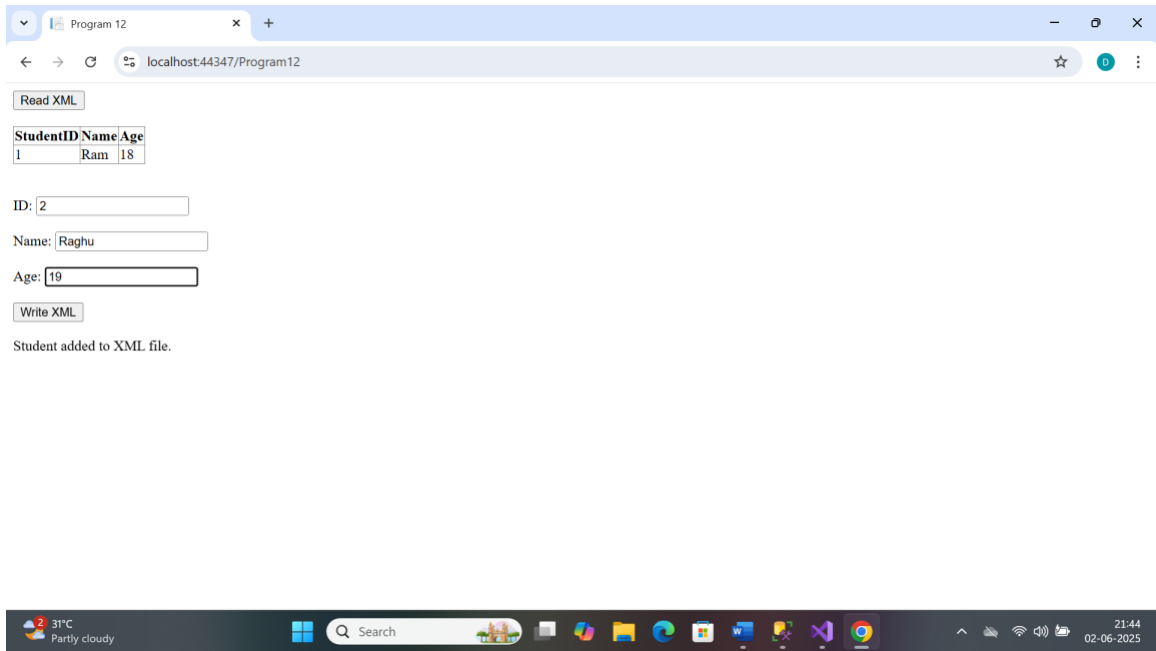
Name:

Age:

Write XML

Student added to XML file.





Program 13: Implement Authentication–Authorization

Demonstrate Forms Authentication with login and protected page.

Program13.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="Program13.aspx.cs"
Inherits="Program13" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
```

```

<head runat="server">
    <title>Login Page</title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <asp:TextBox ID="txtUser" runat="server"
Placeholder="Username"></asp:TextBox><br /><br />
            <asp:TextBox ID="txtPass" runat="server" TextMode="Password"
Placeholder="Password"></asp:TextBox><br /><br />
            <asp:Button ID="btnLogin" runat="server" Text="Login"
OnClick="btnLogin_Click" /><br /><br />
            <asp:Label ID="lblError" runat="server"></asp:Label>
        </div>
    </form>
</body>
</html>

```

Program13.aspx.cs:

```

using System;
using System.Web.Security;

public partial class Login : System.Web.UI.Page
{
    protected void btnLogin_Click(object sender, EventArgs e)
    {
        if (txtUser.Text == "admin" && txtPass.Text == "admin")
        {
            FormsAuthentication.RedirectFromLoginPage(txtUser.Text, false);
            Response.Redirect("Welcome.aspx");
        }
        else
        {
            lblError.Text = "Invalid credentials.";
        }
    }
}

```

Welcome.aspx:

```

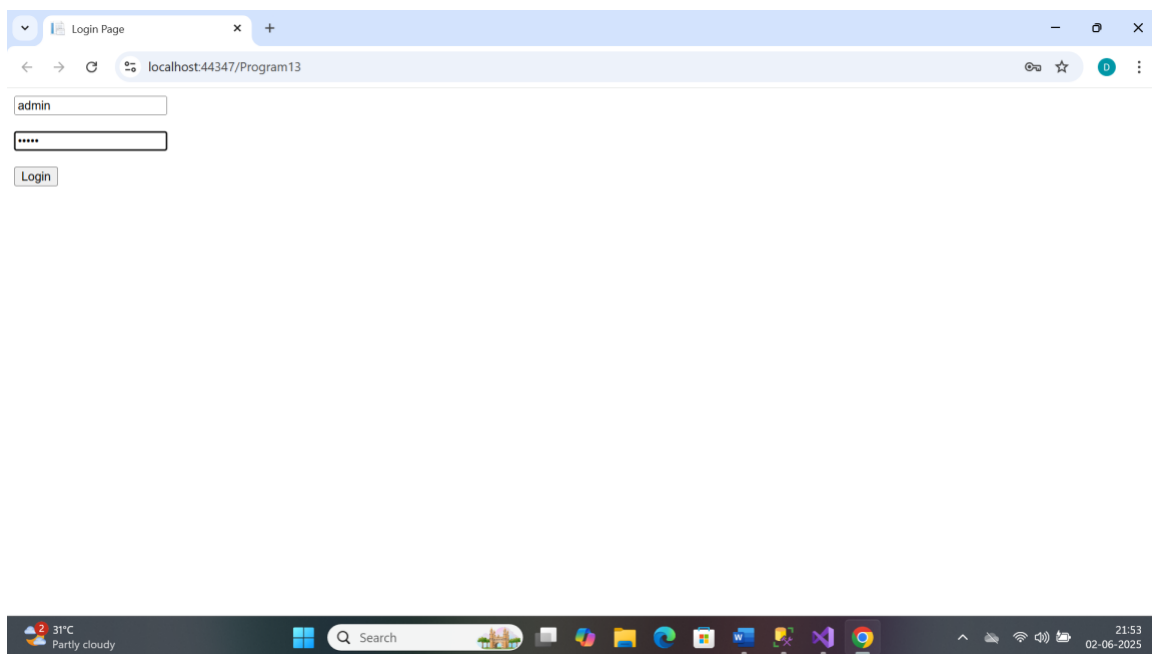
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="Welcome.aspx.cs"
Inherits="Welcome" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title>Welcome</title>
</head>
<body>
    <form id="form1" runat="server">
        <div>

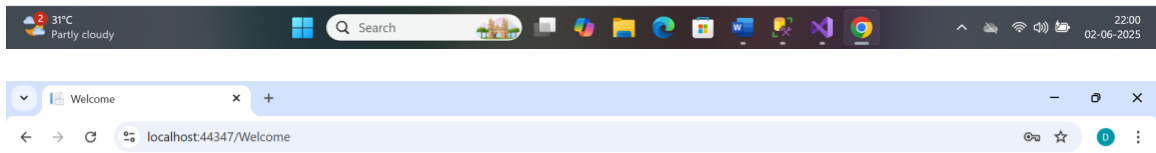
```

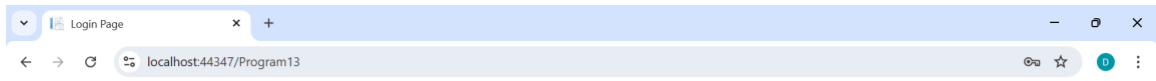
```
<h1> Welcome Admin </h1>
</div>
</form>
</body>
</html>
```

Output:

Program13.aspx authenticates user "admin"/"admin". On success, redirects to Protected.aspx showing "Welcome, admin".







Invalid credentials.



Program 14: Ticket Reservation Using ASP.NET Controls

Create a form to select source, destination, date, and display ticket details.

Program14.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="Program14.aspx.cs"
Inherits="Program14" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title>Ticket Reservation</title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            Source: <asp:DropDownList ID="ddlSource" runat="server">
                <asp:ListItem>City A</asp:ListItem>
                <asp:ListItem>City B</asp:ListItem>
            </asp:DropDownList><br /><br />
            Destination: <asp:DropDownList ID="ddlDestination"
runat="server">
                <asp:ListItem>City X</asp:ListItem>
                <asp:ListItem>City Y</asp:ListItem>
            </asp:DropDownList><br /><br />
            <asp:Calendar ID="Calendar1" runat="server"></asp:Calendar><br
/><br />
            <asp:Button ID="btnReserve" runat="server" Text="Reserve
Ticket" OnClick="btnReserve_Click" /><br /><br />
            <asp:Label ID="lblTicket" runat="server"></asp:Label>
        </div>
    </form>
</body>
</html>
```

Program14.aspx.cs:

```
using System;

public partial class Program14 : System.Web.UI.Page
{
    protected void btnReserve_Click(object sender, EventArgs e)
    {
        lblTicket.Text = "Ticket Reserved: From " +
ddlSource.SelectedItem.Text + " to " + ddlDestination.SelectedItem.Text +
"; Date: " + Calendar1.SelectedDate.ToShortDateString();
    }
}
```

Output:

Displays reservation details: source, destination, and selected date after clicking Reserve Ticket.

Ticket Reservation

localhost:44347/Program14

Source: City A

Destination: City X

June 2025

Sun	Mon	Tue	Wed	Thu	Fri	Sat
25	26	27	28	29	30	31
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	1	2	3	4	5

Reserve Ticket

30°C
Mostly cloudy

Search

22:22
02-06-2025

Ticket Reservation

localhost:44347/Program14

Source: City A

Destination: City X

June 2025

Sun	Mon	Tue	Wed	Thu	Fri	Sat
25	26	27	28	29	30	31
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	1	2	3	4	5

Reserve Ticket

Ticket Reserved: From City A to City X; Date: 28-05-2025

30°C
Mostly cloudy

Search

22:23
02-06-2025

Program 15: Online Examination Using ASP.NET Controls

Create a quiz with multiple choice questions using RadioButtonList and display score.

Program15.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="Program15.aspx.cs"
Inherits="Program15" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">

<head runat="server">

    <title>Online Examination</title>

</head>

<body>

    <form id="form1" runat="server">

        <div>

            <h4>1. What is the capital of India?<br /></h4>

            <asp:RadioButtonList ID="rblQ1" runat="server">

                <asp:ListItem>New Delhi</asp:ListItem>

                <asp:ListItem>Chennai</asp:ListItem>

                <asp:ListItem>Kerala</asp:ListItem>

            </asp:RadioButtonList><br /><br />

            <h4>2. Select programming languages:<br /></h4>

            <asp:CheckBoxList ID="cblQ2" runat="server">

                <asp:ListItem Text="C" Value="C#"></asp:ListItem>

                <asp:ListItem Text="HTML" Value="HTML"></asp:ListItem>

                <asp:ListItem Text="Java" Value="Java"></asp:ListItem>

            </asp:CheckBoxList><br /><br />

            <asp:Button ID="btnSubmitExam" runat="server" Text="Submit Exam"
OnClick="btnSubmitExam_Click" /><br /><br />

            <asp:Label ID="lblScore" runat="server"></asp:Label>

        </div>

    </form>

</body>

</html>
```

```
        </form>
    </body>
</html>
```

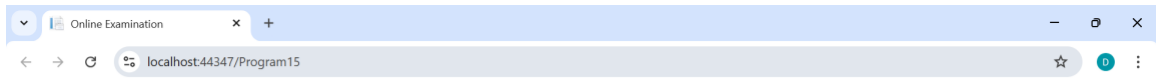
Program15.aspx.cs:

```
using System;
```

```
public partial class Program15 : System.Web.UI.Page
{
    protected void btnSubmitExam_Click(object sender, EventArgs e)
    {
        int score = 0;
        if (rblQ1.SelectedValue == "New Delhi") score += 5;
        foreach (System.Web.UI.WebControls.ListItem li in cblQ2.Items)
        {
            if (li.Selected && (li.Value == "C" || li.Value == "Java")) score
+= 5;
        }
        lblScore.Text = "Your Score: " + score;
    }
}
```

Output:

Displays score: 5 points for each correct answer (Q1 and Q2).



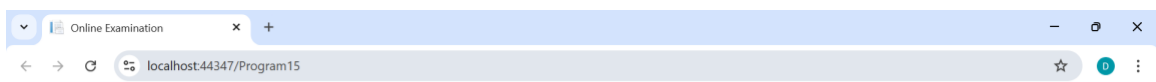
1. What is the capital of India?

- ☐ New Delhi
- ☐ Chennai
- ☐ Kerala

2. Select programming languages:

- ☐ C
- ☐ HTML
- ☐ Java

Submit Exam



1. What is the capital of India?

- ☒ New Delhi
- ☐ Chennai
- ☐ Kerala

2. Select programming languages:

- ☒ C
- ☐ HTML
- ☒ Java

Submit Exam





1. What is the capital of India?

- ☒ New Delhi
- ☐ Chennai
- ☐ Kerala

2. Select programming languages:

- ☒ C
- ☐ HTML
- ☒ Java

[Submit Exam](#)

Your Score: 10



